

ABOUT ELEPHANTVOICES

This [video](#) gives an overview of the incredibly important work done at ElephantVoices.

Our goals are to advance the study of elephant cognition, communication and social behavior, and to promote the scientifically sound and ethical management and care of elephants. We accomplish these through research, conservation, advocacy and the sharing of knowledge.



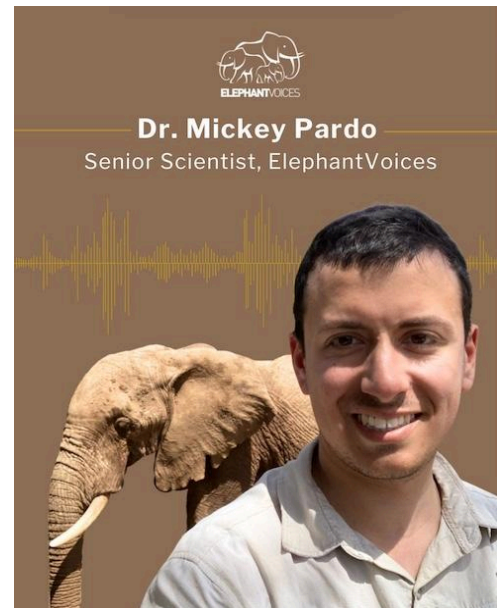
Listening to the voices of elephants over decades has taught us that communication is the glue that binds the social network of an intelligent species, and its study offers a window into the hearts and minds of elephants. Our collection of observations, audio and video recordings, and images form the basis of extensive databases, that are visited and used by a world-wide audience.

A decades-long study of elephant social behaviour, communication and cognition in [Amboseli](#), Kenya, forms the core of our long-term research of these remarkable animals. Our work in the [Maasai Mara](#), Kenya, and [Gorongosa National Park](#), Mozambique, has added to this body of knowledge.

By studying what elephants are capable of understanding and what they communicate to one another, we have a better chance of finding solutions to the many challenges they face. We collaborate with elephant biologists all over the world and our online collections form a unique resource for other scientists and the public.

HACKATHON CHALLENGE

Most of the calls made by elephants are low-frequency vocalizations known as rumbles, which typically have fundamental frequencies between 10-20 Hz, with harmonics often reaching 300-1,000 Hz. One challenge with studying elephant communication is that the frequency range of rumbles overlaps with the low-frequency mechanical noise, such as the sounds produced by airplanes, cars, and generators, which are common in the areas where researchers record elephants. When an elephant rumble overlaps in both time and frequency with these sources of noise, it becomes difficult or impossible to measure the acoustic properties of the elephant rumble without influence from the noise source, rendering these recordings useless for analysis. - **Dr. Mickey Pardo**



Hackathon Prompt:

Develop a method to remove overlapping noise from elephant recordings (airplanes, cars and generators) without distorting the elephant call.

- Use [spectrograms](#) to visualize sound. See [Team Assets](#) for more details.
- [44 audio recordings](#) with 212 elephant calls embedded in mechanical noise
- Some of the calls overlap with each other too.
- [Spreadsheet](#) has sound file name, start time, and end time for each call

Team Direction:

The winning solution will be in the form of a tool, platform and/or trained model that can remove background noise and accurately isolate the largest number of low-frequency elephant vocalizations from these recordings. Each team should focus on completing the process of isolating an elephant rumble from a minimum of one recording in one category (airplane, car or generator). For this hackathon, quality beats quantity but bonus points will be awarded for teams that are able to complete the process for multiple recordings and/or categories. See the [Q&A](#) section for more details about the process.

Team Assets:

- **Audio:** We have provided you with 44 recordings, in this [shared folder](#), that include at least one elephant rumble overlapping with noise from an airplane, car, or generator and a [spreadsheet](#) indicating the start and end time of the rumble(s) in each recording. All audio files listed in the spreadsheet have full audio spectrum (0Hz and above)
- **Visual:** We have provided you with spectrograms of the recordings in this [shared folder](#) so you can see the entire range for your work.
- **Tool:** Download [Audacity](#).
 - Open sound file
 - File > Open (or drag and drop)
 - Switch to spectrogram view
 - Drop down menu top left corner of sound > Spectrogram
 - Adjust spectrogram parameters:
 - Edit > Preferences > Tracks > Spectrograms
 - Change window size to adjust resolution
 - Longer window = better frequency resolution, worse time resolution
- **Workshop:** Dr Mickey Pardo will be walking through this <presentation> to provide more details about the project and how you can fully leverage the assets provided.

Q&A

Why do we want clean recordings?

- To decipher the meanings of elephant calls, biologists and linguists need to be able to measure the acoustic properties of those calls without interference from other overlapping sound sources. Many recordings of elephant calls collected in the wild overlap with either calls from other elephants, outside sources of noise (cars, planes, generators, etc.), or both, rendering them unusable for analysis. Developing a method to remove these sources of noise will allow biologists and linguists to make use of a larger portion of the existing recordings of wild elephants, improving their ability to decipher how elephants communicate with each other.

What will these recordings be used for?

- Isolated (cleaned) recordings of elephant calls will be added to a larger data base (and potentially AI model) that can be leveraged to analyze the language of elephants to understand what they are saying to each other.

Is there anything special about the recordings provided for this hackathon?

- Yes, most elephant calls are a type of vocalization known as “rumbles”, which contain frequencies that humans cannot hear. Therefore, you will need to rely on a visual representation of sound known as a spectrogram rather than on listening to the recordings.
- Like most natural sounds, elephant rumbles contain multiple frequencies at the same time within a single call. In the spectrogram, these frequencies look like horizontal bands stacked on top of each other. The fundamental (lowest) frequency is typically between 10-20 Hz. Then there are harmonics layered on top of the fundamental frequency, which are whole-integer multiples of it (2x the fundamental, 3x the fundamental, etc.). When you listen to a recording of a rumble, what you are hearing is actually just the higher harmonics of the call. That is because the fundamental frequency is below what the human ear is capable of hearing, and even the lower harmonics are below what most loudspeakers can play.
- The highest harmonic in an elephant rumble may be as high as 1000 Hz (50-100 times the fundamental frequency). However, higher frequencies typically don't travel as far as lower frequencies, so the further away the elephant was from the microphone when she was recorded, the less visible the higher harmonics will be in



a spectrogram. For rumbles that were recorded from very far away, you may only see the fundamental frequency and the next lowest harmonic, and you likely will not be able to hear this call at all when you play it back.

- Another consideration is that elephants talk over each other just like humans. You will need to consider multiple elephant speakers. You can identify when multiple elephants are calling on top of each other by looking at the spectrogram. If there are two different callers, the harmonics of the two calls will often cross. Within a single call, the harmonics will never cross each other.

OTHER RESOURCES

Below is a list of some current AI tools that provide noise and sound removal at different frequencies that you can consider in your work.

- <https://veed.io>
- <https://media.io>
- <https://weet.co>
- <https://podcastle.ai>
- <https://lalal.ai>
- <https://kapwing.com>
- <https://products.aspose.app>
- <https://rgb.d.com>

